The Æther-flow Interpretation of Relativity

Ordered-Flow Label Symmetry/Quotient Candidate: Architecture Screen

Abstract

The explicit ordered-flow label symmetry/quotient candidate has already passed the flat-background coefficient-level screen: the observer-equation correction tensor vanishes while the bridge metric and the single-metric matter route remain fixed. The next burden is narrower and genuinely nonlinear. One must decide whether that quotient is an honest branch-level redundancy or only a bookkeeping trick that hides the old infinite-dimensional fiber in another form.

This note performs exactly that architecture screen and stops there. Start from the same tuned double exact-square candidate with

$$\mathcal{A}_Y \equiv \frac{\sqrt{\lambda_4}}{M_*}\Delta_\perp S - M_Y Y, \qquad \mathcal{A}_Z \equiv m_S\big(U^A\nabla_A S - \sigma_0\big) - M_Z Z,$$

and the same local ordered-flow relabeling symmetry

$$S \mapsto S + \eta, \qquad Y \mapsto Y + \frac{\sqrt{\lambda_4}}{M_Y M_*}\Delta_\perp \eta, \qquad Z \mapsto Z + \frac{m_S}{M_Z}U^A\nabla_A\eta,$$

with G_{AB}, U^A, Q^I, and the bridge metric fixed. The first result is structural: the quotient field space is identified cleanly and globally. For any fixed reference ordered-flow label S_* in the chosen asymptotic class, every $\mathscr{G}_{\mathrm{of}}$-orbit contains a unique representative with $S = S_*$. Equivalently,

$$\mathscr{F}_{\mathrm{phys}} \cong \big\{\big(G_{AB}, U^A, Q^I, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z\big)\big\},$$

so the quotient has an explicit physical coordinate description.

The second result is branch-level. The action descends exactly to the quotient action

$$S^{\mathrm{phys}}_{4YZ} = \frac{c^3}{16\pi G_N}\int_{\mathcal{A}} d^4X\,\sqrt{G}\,\big(R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}\big) + \int_{\mathcal{A}} d^4X\,\sqrt{G}\,\mathcal{L}_{\mathrm{aux,phys}} + S_{\mathrm{matter}}[\Xi^*\mathfrak{g}, \psi],$$

with

$$\mathcal{L}_{\mathrm{aux,phys}} = \lambda_U(G_{AB}U^AU^B - 1) - \frac{1}{2}\mathcal{A}_Y^2 - \frac{1}{2}\mathcal{A}_Z^2 - \frac{1}{2}\delta_{IJ}P^{AB}(D_AQ^I)(D_BQ^J) - \frac{1}{2}\widehat{M}_{IJ}Q^IQ^J - \frac{\kappa_\theta}{2}(\nabla_AU^A)^2 - \frac{\kappa_\Omega}{4}\Omega_A$$

Its quotient Euler–Lagrange equations are $\mathcal{A}_Y = 0$, $\mathcal{A}_Z = 0$, together with the reduced S-free equations already encoded in the old degenerate action. On shell, the physical branch data are therefore just the reduced fields

$$\big(G_{AB}, U^A, Q^I, \lambda_U, \psi\big)$$

modulo the already recorded geometric gauge choices. The old infinite-dimensional substrate fiber survives only as the family of gauge representatives of a single physical state.

The architecture verdict is therefore positive at current disciplined scope. The ordered-flow relabeling is an honest physical redundancy of the branch, not merely a flat-background coefficient trick, and no revived fiber ambiguity survives after quotienting. This note still stops before any reduced-system, local-coercive, or theorem reopening. It records only that such a reopening is no longer blocked by the old branch-level underdetermination.

1. Introduction

The explicit ordered-flow symmetry/quotient manuscript settled one task and left one sharper one. It showed that the doubly tuned exact-square branch can be reinterpreted by an exact local ordered-flow relabeling symmetry and that, at flat-background quadratic scope, the observer-equation correction tensor vanishes on that quotient branch. What it did not yet decide was whether that symmetry remains honest at full branch level.

That is the immediate burden now. The present note does not ask for a reduced $3+1$ system, a local theorem, or a coercive estimate. It asks only whether the quotient architecture is real. Concretely, three issues must be settled:

1. what the physical field space actually is after quotienting by the ordered-flow relabeling symmetry;
2. whether the old infinite-dimensional substrate fiber becomes pure gauge rather than surviving as hidden physical ambiguity;
3. whether the quotient candidate therefore survives the branch-level architecture screen while the bridge metric and single-metric matter route remain fixed.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present manuscript stays strictly inside that boundary. It does not claim a completed substrate derivation. It does not prove local well-posedness or slice closure. It performs only the quotient-architecture screen and stops at the corresponding verdict.

2. The Branch To Be Screened

Definition 1 (Representative field space)**.** *Let $\mathscr{F}_{\mathrm{rep}}$ denote the space of field tuples*

$$(G_{AB}, U^A, S, Q^I, Y, Z, \lambda_U, \psi)$$

obeying the same positivity assumptions, admissibility assumptions, and asymptotic ordered-flow profile as in the explicit quotient-candidate note. On $\mathscr{F}_{\rm rep}$, define

$$\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B, \qquad P_{AB} = G_{AB} - U_A U_B, \tag{1}$$

$$\Delta_\perp S \equiv \nabla_A (P^{AB} \nabla_B S), \qquad \Omega_{AB} \equiv P_A{}^C P_B{}^D \nabla_{[C} U_{D]}, \tag{2}$$

and

$$\mathcal{A}_Y \equiv \frac{\sqrt{\lambda_4}}{M_*} \Delta_\perp S - M_Y Y, \qquad \mathcal{A}_Z \equiv m_S (U^A \nabla_A S - \sigma_0) - M_Z Z. \tag{3}$$

Definition 2 (Representative action)**.** *The representative action on $\mathscr{F}_{\rm rep}$ is*

$$S^{\rm rep}_{4YZ} = \frac{c^3}{16\pi G_N} \int_{\mathcal{A}} d^4X \sqrt{G}\, (R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}) + \int_{\mathcal{A}} d^4X \sqrt{G}\, \mathcal{L}_{{\rm aux},4YZ} + S_{\rm matter}[\Xi^* \mathfrak{g}, \psi], \tag{4}$$

with

$$\begin{aligned} \mathcal{L}_{{\rm aux},4YZ} = {} & \lambda_U (G_{AB} U^A U^B - 1) - \frac{1}{2} \mathcal{A}_Y^2 - \frac{1}{2} \mathcal{A}_Z^2 \\ & - \frac{1}{2} \delta_{IJ} P^{AB} (D_A Q^I)(D_B Q^J) - \frac{1}{2} \widehat{M}_{IJ} Q^I Q^J - \frac{\kappa_\theta}{2} (\nabla_A U^A)^2 - \frac{\kappa_\Omega}{4} \Omega_{AB} \Omega^{AB}, \end{aligned} \tag{5}$$

where

$$m_S^2 > 0, \qquad \lambda_4 > 0, \qquad M_Y^2 > 0, \qquad M_Z^2 > 0, \qquad \widehat{M}_{IJ} > 0, \qquad \kappa_\theta > 0, \qquad \kappa_\Omega > 0. \tag{6}$$

Definition 3 (Ordered-flow relabeling group)**.** *Let $\mathscr{G}_{\rm of}$ denote the local transformation group generated by smooth parameters η preserving the chosen asymptotic ordered-flow profile. Its action on $\mathscr{F}_{\rm rep}$ is*

$$\delta_\eta S = \eta, \qquad \delta_\eta Y = \frac{\sqrt{\lambda_4}}{M_Y M_*} \Delta_\perp \eta, \qquad \delta_\eta Z = \frac{m_S}{M_Z} U^A \nabla_A \eta, \tag{7}$$

with

$$\delta_\eta G_{AB} = 0, \qquad \delta_\eta U^A = 0, \qquad \delta_\eta Q^I = 0, \qquad \delta_\eta \lambda_U = 0, \qquad \delta_\eta \psi = 0. \tag{8}$$

Proposition 1 (Exact invariance)**.** *The representative action $S^{\rm rep}_{4YZ}$ is exactly invariant under $\mathscr{G}_{\rm of}$. In particular, $\mathcal{A}_Y$, $\mathcal{A}_Z$, $\mathfrak{g}_{AB}$, the observer metric $g_{\mu\nu} = \Xi^* \mathfrak{g}_{AB}$, and the matter route are all invariant.*

Proof. Equation (7) gives

$$\delta_\eta (\Delta_\perp S) = \Delta_\perp \eta, \qquad \delta_\eta (U^A \nabla_A S) = U^A \nabla_A \eta,$$

because G_{AB} and U^A are fixed under the relabeling. Substituting into (3) gives $\delta_\eta \mathcal{A}_Y = 0$ and $\delta_\eta \mathcal{A}_Z = 0$. The bridge metric depends only on G_{AB} and U^A, so it is invariant as well. Every term of (4)–(5) therefore depends only on invariant quantities. □

3. Physical Field Space After Quotienting

Definition 4 (Current quotient-architecture criterion)**.** *At current disciplined scope, the ordered-flow quotient candidate passes its architecture screen if all three statements hold:*

1. *the physical field space after quotienting is identified explicitly and without leftover representative ambiguity;*

2. *the action descends to that quotient field space as a well-defined physical action;*

3. *on shell, the old infinite-dimensional substrate fiber is exactly a gauge orbit rather than a surviving physical underdetermination.*

This criterion concerns branch-level physical data only. It does not yet ask for a reduced-system, local-coercive, or theorem package.

Proposition 2 (Orbit invariants classify representatives)**.** *Define the orbit map*

$$\pi : \mathscr{F}_{\mathrm{rep}} \to \left\{ \left(G_{AB}, U^A, Q^I, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z \right) \right\} \tag{9}$$

by

$$\pi(G, U, S, Q, Y, Z, \lambda_U, \psi) = (G, U, Q, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z).$$

Then:

1. *π is constant on $\mathscr{G}_{\mathrm{of}}$-orbits;*

2. *two representatives have the same image under π if and only if they are related by an ordered-flow relabeling.*

Proof. Item (1) is Proposition 1. For item (2), suppose first that the two representatives are gauge related. Then they have the same G_{AB}, U^A, Q^I, λ_U, ψ, $\mathcal{A}_Y$, and $\mathcal{A}_Z$ by construction.

Conversely, suppose

$$\pi(G, U, S_1, Q, Y_1, Z_1, \lambda_U, \psi) = \pi(G, U, S_2, Q, Y_2, Z_2, \lambda_U, \psi).$$

Then G, U, Q, λ_U, and ψ already agree, and equality of $\mathcal{A}_Y$ and $\mathcal{A}_Z$ gives

$$M_Y(Y_2 - Y_1) = \frac{\sqrt{\lambda_4}}{M_*} \Delta_\perp (S_2 - S_1),$$

and

$$M_Z(Z_2 - Z_1) = m_S U^A \nabla_A (S_2 - S_1).$$

Setting $\eta = S_2 - S_1$, these are exactly the transformation laws (7). Hence the two representatives lie on the same $\mathscr{G}_{\mathrm{of}}$-orbit. □

Corollary 1 (Unique reference slice)**.** *Fix any smooth scalar field S_* in the chosen asymptotic ordered-flow class, and define*

$$\Sigma_* \equiv \left\{ (G, U, S, Q, Y, Z, \lambda_U, \psi) \in \mathscr{F}_{\mathrm{rep}} : \ S = S_* \right\}. \tag{10}$$

Then every $\mathscr{G}_{\mathrm{of}}$-orbit intersects Σ_* in exactly one point.

Proof. Given any representative with ordered-flow label S, choose $\eta = S_* - S$. Since S and S_* obey the same asymptotic ordered-flow condition, this η belongs to the allowed relabeling class. The transformed representative has $S = S_*$, so every orbit meets Σ_*.

If two points of the same orbit lie in Σ_*, then their ordered-flow labels both equal S_*. But $\delta_\eta S = \eta$, so the only relabeling preserving S_* is $\eta = 0$. Therefore the intersection is unique. □

Definition 5 (Physical field space)**.** *The physical field space of the quotient branch is*

$$\mathscr{F}_{\text{phys}} \equiv \mathscr{F}_{\text{rep}}/\mathscr{G}_{\text{of}}. \tag{11}$$

By Proposition 2 and Corollary 1, it can be represented equivalently either by the unique slice Σ_* or by the invariant-coordinate space

$$\mathscr{F}_{\text{phys}} \cong \left\{ (G_{AB}, U^A, Q^I, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z) \right\}. \tag{12}$$

Remark 1. *Definition 5 is the first new architecture result beyond the coefficient-level note. The quotient is not an abstract slogan. It has both a unique representative model $S = S_*$ and a direct invariant-coordinate model $(G, U, Q, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z)$.*

4. The Quotient Action

Theorem 1 (Action descends exactly)**.** *There exists a unique action S^{phys}_{4YZ} on $\mathscr{F}_{\text{phys}}$ whose pullback to $\mathscr{F}_{\text{rep}}$ is S^{rep}_{4YZ}. In invariant coordinates* (12)*, it is*

$$S^{\text{phys}}_{4YZ} = \frac{c^3}{16\pi G_N} \int_{\mathcal{A}} d^4X \sqrt{G}\, (R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}) + \int_{\mathcal{A}} d^4X \sqrt{G}\, \mathcal{L}_{\text{aux,phys}} + S_{\text{matter}}[\Xi^*\mathfrak{g}, \psi], \tag{13}$$

with

$$\begin{aligned} \mathcal{L}_{\text{aux,phys}} = {} & \lambda_U (G_{AB} U^A U^B - 1) - \frac{1}{2}\mathcal{A}_Y^2 - \frac{1}{2}\mathcal{A}_Z^2 \\ & - \frac{1}{2}\delta_{IJ} P^{AB} (D_A Q^I)(D_B Q^J) - \frac{1}{2}\widehat{M}_{IJ} Q^I Q^J - \frac{\kappa_\theta}{2}(\nabla_A U^A)^2 - \frac{\kappa_\Omega}{4}\Omega_{AB}\Omega^{AB}. \end{aligned} \tag{14}$$

Proof. Proposition 1 shows that S^{rep}_{4YZ} is constant on every $\mathscr{G}_{\text{of}}$-orbit. Therefore it descends uniquely to the orbit space $\mathscr{F}_{\text{phys}}$. Formula (13)–(14) is exactly the representative action rewritten in orbit coordinates, because every appearance of S, Y, and Z in (4)–(5) is through the invariant combinations $\mathcal{A}_Y$ and $\mathcal{A}_Z$. □

Proposition 3 (Quotient equations for the ordered-flow sector)**.** *The quotient Euler–Lagrange equations obtained by varying $\mathcal{A}_Y$ and $\mathcal{A}_Z$ in S^{phys}_{4YZ} are*

$$\mathcal{A}_Y = 0, \qquad \mathcal{A}_Z = 0. \tag{15}$$

Proof. In (14), the invariant fields $\mathcal{A}_Y$ and $\mathcal{A}_Z$ appear only through the algebraic exact squares $-\frac{1}{2}\mathcal{A}_Y^2$ and $-\frac{1}{2}\mathcal{A}_Z^2$. Varying them gives (15). □

Corollary 2 (On-shell physical branch data)**.** *On shell, the quotient branch is represented by*

$$(G_{AB}, U^A, Q^I, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z) \quad \textit{with} \quad \mathcal{A}_Y = \mathcal{A}_Z = 0. \tag{16}$$

The remaining equations are exactly those obtained from the S-free reduced action

$$S_{\rm deg} = \frac{c^3}{16\pi G_N}\int_{\mathcal{A}} d^4X\,\sqrt{G}\,\big(R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}\big) + \int_{\mathcal{A}} d^4X\,\sqrt{G}\,\mathcal{L}_{\rm aux,deg} + S_{\rm matter}[\Xi^*\mathfrak{g},\psi], \tag{17}$$

with

$$\begin{aligned}\mathcal{L}_{\rm aux,deg} = \lambda_U\big(G_{AB}U^AU^B - 1\big) - \frac{1}{2}\delta_{IJ}P^{AB}(D_AQ^I)(D_BQ^J) - \frac{1}{2}\widehat{M}_{IJ}Q^IQ^J \\ - \frac{\kappa_\theta}{2}\big(\nabla_AU^A\big)^2 - \frac{\kappa_\Omega}{4}\Omega_{AB}\Omega^{AB}.\end{aligned} \tag{18}$$

Proof. Proposition 3 gives $\mathcal{A}_Y = \mathcal{A}_Z = 0$. After imposing those two equations, the remaining terms of (13)–(14) are precisely (17)–(18). Hence the remaining quotient equations are exactly the Euler–Lagrange equations of $S_{\rm deg}$. □

Remark 2. *Corollary 2 is the second essential architecture result. The old degenerate action no longer signals physical underdetermination, because the missing S-equation belongs to a field that is no longer physical on the quotient branch. What remains physically is the reduced S-free system together with the algebraic invariant conditions $\mathcal{A}_Y = \mathcal{A}_Z = 0$.*

5. No Revived Fiber Ambiguity

Theorem 2 (The old infinite-dimensional fiber is pure gauge)**.** *Fix any on-shell quotient configuration $(G, U, Q, \lambda_U, \psi, \mathcal{A}_Y, \mathcal{A}_Z)$ with $\mathcal{A}_Y = \mathcal{A}_Z = 0$. The full representative configurations projecting to it form exactly one $\mathscr{G}_{\rm of}$-orbit. Equivalently, the old infinite-dimensional family of lifts indexed by S is not a family of distinct physical states.*

Proof. By Definition 5, a representative projects to the given quotient point precisely when it has the same G, U, Q, λ_U, ψ, $\mathcal{A}_Y$, and $\mathcal{A}_Z$. Proposition 2 then says that any two such representatives are gauge related. Since the old lift family differed only by choices of (S, Y, Z) holding G, U, Q, λ_U, ψ, $\mathcal{A}_Y$, and $\mathcal{A}_Z$ fixed, it is exactly one gauge orbit. □

Corollary 3 (No revived physical underdetermination)**.** *After quotienting, the branch has no additional physical ambiguity associated with the ordered-flow label. The physical data of the on-shell branch are*

$$(G_{AB}, U^A, Q^I, \lambda_U, \psi) \tag{19}$$

modulo the already recorded geometric gauge choices, and distinct representatives (S, Y, Z) of the same data are gauge copies only.

Proof. Corollary 2 removes $\mathcal{A}_Y$ and $\mathcal{A}_Z$ on shell. Theorem 2 then shows that any residual freedom in S, Y, and Z is pure gauge rather than a new physical fiber. □

Remark 3. *Corollary 3 is the exact answer to the question posed by the present manuscript. The quotient does not merely rename the old underdetermined branch. It removes the ordered-flow label from the physical state space.*

6. Architecture Verdict and Stopping Point

Theorem 3 (Positive quotient-architecture verdict)**.** *The explicit ordered-flow symmetry/quotient candidate passes the current branch-level architecture screen while the bridge metric and single-metric matter route remain fixed. More precisely:*

1. *the physical field space after quotienting is identified explicitly by Definition 5;*
2. *the action descends exactly to the quotient action S^{phys}_{4YZ} of Theorem 1;*
3. *the old infinite-dimensional substrate fiber is pure gauge by Theorem 2, so no revived physical ambiguity survives modulo the ordered-flow relabeling symmetry.*

Hence the ordered-flow relabeling is an honest physical redundancy of the branch, not merely a coefficient-level bookkeeping device.

Proof. Item (1) is Proposition 2 together with Corollary 1. Item (2) is Theorem 1. Item (3) is Theorem 2 and Corollary 3. These are exactly the three items required by Definition 4. □

Corollary 4 (What this note does and does not justify)**.** *The present note justifies exactly two programmatic conclusions:*

1. *the quotient candidate survives honestly beyond the flat-background coefficient-level screen as a nonlinear branch-level architecture;*
2. *any future reduced-system reopening on this quotient branch is no longer blocked by the old ordered-flow fiber ambiguity.*

It does not *yet justify a reduced-system note, a local-coercive note, or any theorem claim on that branch.*

Proof. Item (1) is Theorem 3. Item (2) follows because the branch-level ambiguity tested here was the only specific obstruction inherited from the old double exact-square architecture note once the quotient principle had been introduced. The last sentence is exactly the stopping rule built into Definition 4. □

Remark 4. *If the present architecture screen had failed while the bridge metric and single-metric matter route remained fixed, the next larger move would have been to reconsider one of those two ingredients. Theorem 3 shows that this contingency is not the current verdict. The next question, if the active line chooses to proceed, is whether a reduced-system note should now be reopened on the quotient branch.*

7. Conclusion

The present manuscript performs the first nonlinear screen after the coefficient-level positive quotient verdict and stops exactly where it should.

1. It identifies the true physical field space after quotienting by the ordered-flow relabeling symmetry.
2. It shows that the quotient action is well defined and that its ordered-flow-sector equations are simply $\mathcal{A}_Y = \mathcal{A}_Z = 0$.

3. It proves that the old infinite-dimensional substrate fiber is exactly one gauge orbit and therefore no longer a physical underdetermination.

4. It records the resulting architecture verdict: the quotient candidate is honest at branch level while the bridge metric and single-metric matter route remain fixed.

That is the correct stopping point. The quotient branch has passed the architecture screen, but no reduced-system, local-coercive, or theorem work is reopened here. This note records only that the old fiber ambiguity no longer blocks such a reopening in principle.